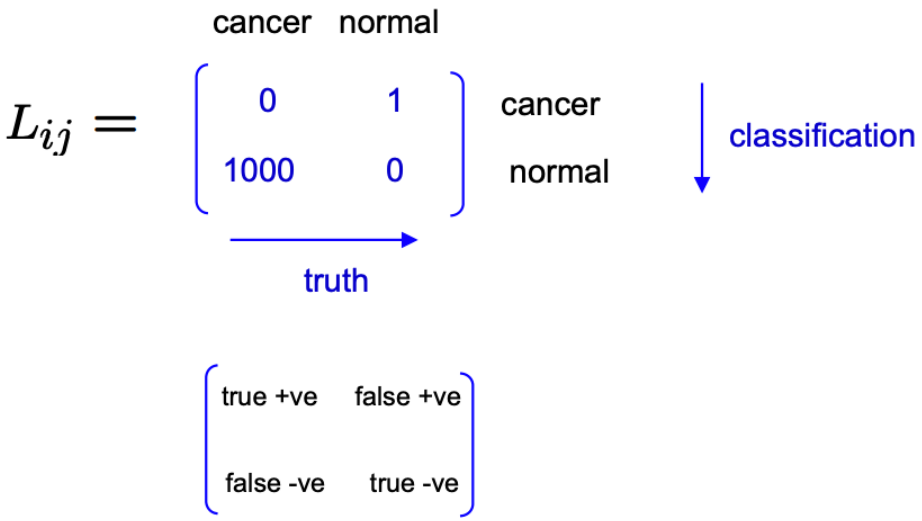
Incorporating Risk Function

Bayesian Decision Theory – Risk Function

Loss matrix



- Loss function
  - sea bass: 600 per kg
  - Salmon: 300 per kg

$$L = \begin{bmatrix} 0 & -300 \\ +300 & 0 \end{bmatrix} \begin{matrix} \text{Seabass} \\ \text{Salmon} \end{matrix}$$

Seabass Salmon

Zero-one loss function

$$L = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Bayesian Decision Theory – Risk Function

- Let $\{\omega 1, \omega 2, \ldots, \omega c\}$ be the set of c states of nature (or "categories")
- Let $\{\alpha 1, \alpha 2, \ldots, \alpha a\}$ be the set of possible actions
- Let $\lambda(\alpha i \mid \omega j)$ be the loss incurred for taking action αi when the true state of nature is ωj

Two-category Classification

- Actions:
  - $\alpha 1$: deciding $\omega 1$
  - $\alpha 2$: deciding $\omega 2$

- $\lambda ij = \lambda(\alpha i \mid \omega j)$
  - Loss incurred for deciding αi when the true state of nature is ωj

Two-category Classification

- $\alpha 1$: deciding $\omega 1$
- $\alpha 2$: deciding $\omega 2$
- $\lambda ij = \lambda(\alpha i \mid \omega j)$
- Loss incurred for deciding αi when the true state of nature is ωj
- Conditional risk:

$$R(\alpha_1|\mathbf{x}) = \lambda_{11}P(\omega_1|\mathbf{x}) + \lambda_{12}P(\omega_2|\mathbf{x})$$
$$R(\alpha_2|\mathbf{x}) = \lambda_{21}P(\omega_1|\mathbf{x}) + \lambda_{22}P(\omega_2|\mathbf{x}).$$

Two-category Classification

- Our rule is the following:

  if R(α1 | x) < R(α2 | x)
- Action α1: “decide ω1” is taken
- This results in the equivalent rule :
- Decide ω1 if:

$$(\lambda_{21} - \lambda_{11})p(\mathbf{x}|\omega_1)P(\omega_1) > (\lambda_{12} - \lambda_{22})p(\mathbf{x}|\omega_2)P(\omega_2)$$

- and decide ω2 otherwise

Bayesian Decision Theory – Continuous Features…

- Overall risk

  R = Sum of all $R(\alpha i \mid x)$ for i = 1,…,a

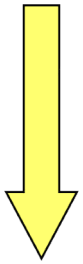
  Conditional risk

- Minimizing R ⟺ Minimizing $R(\alpha i \mid x)$ for i = 1,…, a

- for i = 1,…,a

Bayesian Decision Theory – Continuous Features…

- Select the action αi for which $R(\alpha i \mid x)$ is minimum



- R is minimum and is called the Bayes risk = best performance that can be achieved!

Likelihood Ratio

$$(\lambda_{21} - \lambda_{11})p(\mathbf{x}|\omega_1)P(\omega_1) > (\lambda_{12} - \lambda_{22})p(\mathbf{x}|\omega_2)P(\omega_2)$$

- The preceding rule is equivalent to the following rule:
- If $$\frac{p(\mathbf{x}|\omega_1)}{p(\mathbf{x}|\omega_2)} > \frac{\lambda_{12} - \lambda_{22}}{\lambda_{21} - \lambda_{11}} \frac{P(\omega_2)}{P(\omega_1)}$$

- Then take action $\alpha 1$ (decide $\omega 1$)
- Otherwise take action $\alpha 2$ (decide $\omega 2$)

Likelihood Ratio…

- Optimal decision property

  “If the likelihood ratio exceeds a threshold value independent of the input pattern x, we can take optimal actions”

Example: Checking on a course

- A student needs to make a decision which courses to take, based only on first lecture's impression
- From student's previous experience:

Quality of the course	good	fair	bad
Probability (prior)	0.2	0.4	0.4

- These are **prior probabilities**.

Example: Checking on a course

- The student also knows the class-conditionals:

$Pr(x/w_j)$	good	fair	bad
Interesting lecture	0.8	0.5	0.1
Boring lecture	0.2	0.5	0.9

- The loss function is given by the matrix

$\lambda(\alpha i \mid \omega j)$	good course	fair course	bad course
Taking the course	0	5	10
Not taking the course	20	5	0

Example: Checking on a course

- We can take decisions according to two rules:
  - Posterior probability
  - Risk minimization

Example: Checking on a course

- Given that the first class was **interesting**, the student wants to make an optimal decision.
  - Pr(bad|interesting)
  - Pr(fair|interesting)
  - Pr(good|interesting)

  - Pr(interesting|bad)
  - Pr(interesting|fair)
  - Pr(interesting|good)

  - Pr(interesting)

Example: Checking on a course

- The probability to get the "interesting lecture"(x= interesting):
- Pr(interesting)= Pr(interesting|good course)* Pr(good course)
  + Pr(interesting|fair course)* Pr(fair course)
  + Pr(interesting|bad course)* Pr(bad course)
  =0.8*0.2+0.5*0.4+0.1*0.4=0.4

- Consequently, Pr(boring)=1-0.4=0.6
- Suppose the lecture was interesting. Then we want to compute the **posterior** probabilities of each one of the 3 possible "states of nature".

Example: Checking on a course

$$\Pr(\text{good course}|\text{interesting lecture})$$
$$= \frac{\Pr(\text{interesting}|\text{good})\Pr(\text{good})}{\Pr(\text{interesting})} = \frac{0.8*0.2}{0.4} = 0.4$$

$$\Pr(\text{fair}|\text{interesting})$$
$$= \frac{\Pr(\text{interesting}|\text{fair})\Pr(\text{fair})}{\Pr(\text{interesting})} = \frac{0.5*0.4}{0.4} = 0.5$$

- We can get Pr(bad|interesting) = 0.1 either by the same method, or by noting that it complements the above two to 1.
- Now, we have all we need to make an intelligent decision about an optimal action

Example: Checking on a course

- The student needs to minimize the conditional risk

$$R(\alpha_i \mid x) = \sum_{j=1}^{c} \lambda(\alpha_i \mid \omega_j) P(\omega_j \mid x)$$

R(taking|interesting) =

Pr(good|interesting) λ(taking good course) +
Pr(fair|interesting) λ(taking fair course) +

Pr(bad|interesting) λ(taking bad course)

= 0.4*0 + 0.5*5 + 0.1*10 = 3.5

Example: Checking on a course

$$R(\alpha_i \mid x) = \sum_{j=1}^{c} \lambda(\alpha_i \mid \omega_j) P(\omega_j \mid x)$$

R(not taking|interesting) =

Pr(good|interesting)λ(not taking good course) +
Pr(fair|interesting)λ(not taking fair course) +
Pr(bad|interesting)λ(not taking bad course)

= 0.4*20+0.5*5+0.1*0 = 10.5

Constructing an optimal decision function

- So, if the first lecture was interesting, the student will minimize the conditional risk by taking the course.
- In order to construct the full decision function, we need to define the risk minimization action for the case of boring lecture, as well.

Exercise

- Select the optimal decision where: W = {ω1, ω2}
- P(x | ω1) ⇨ N(2, 0.5) (Normal distribution)
- P(x | ω2) ⇨ N(1.5, 0.2)

- P(ω1) = 2/3
- P(ω2) = 1/3

$$\lambda = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Questions?

Discriminant Functions for the Normal Density…

- Case $\Sigma_i = \Sigma$ (covariance of all classes are identical but arbitrary!)

$$g_i(x) = -\frac{1}{2}(x-\mu_i)^t \Sigma_i^{-1}(x-\mu_i) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma_i| + \ln P(\omega_i)$$

  - Expand the term and disregard the quadratic expression

where :

$$g_i(x) = w_i^t x + w_{i0} \qquad w_i = \Sigma^{-1}\mu; \;\; w_{i0} = -\frac{1}{2}\mu_i^t \Sigma^{-1}\mu_i + \ln P(\omega_i)$$

Discriminant Functions for the Normal Density…

$$x_0 = \frac{1}{2}(\mu_i + \mu_j) - \frac{\ln\left[P(\omega_i)/P(\omega_j)\right]}{(\mu_i - \mu_j)^t \Sigma^{-1}(\mu_i - \mu_j)}.(\mu_i - \mu_j)$$

- Comments about this hyperplane:
  - It passes through $\mathbf{x_0}$
  - It is NOT orthogonal to the line linking the means.
  - What happens when $P(\omega_i) = P(\omega_j)$?
  - If $P(\omega_i) \mathrel{!}= P(\omega_j)$, then $\mathbf{x_0}$ shifts away from the more likely mean.

Discriminant Functions for the Normal Density…

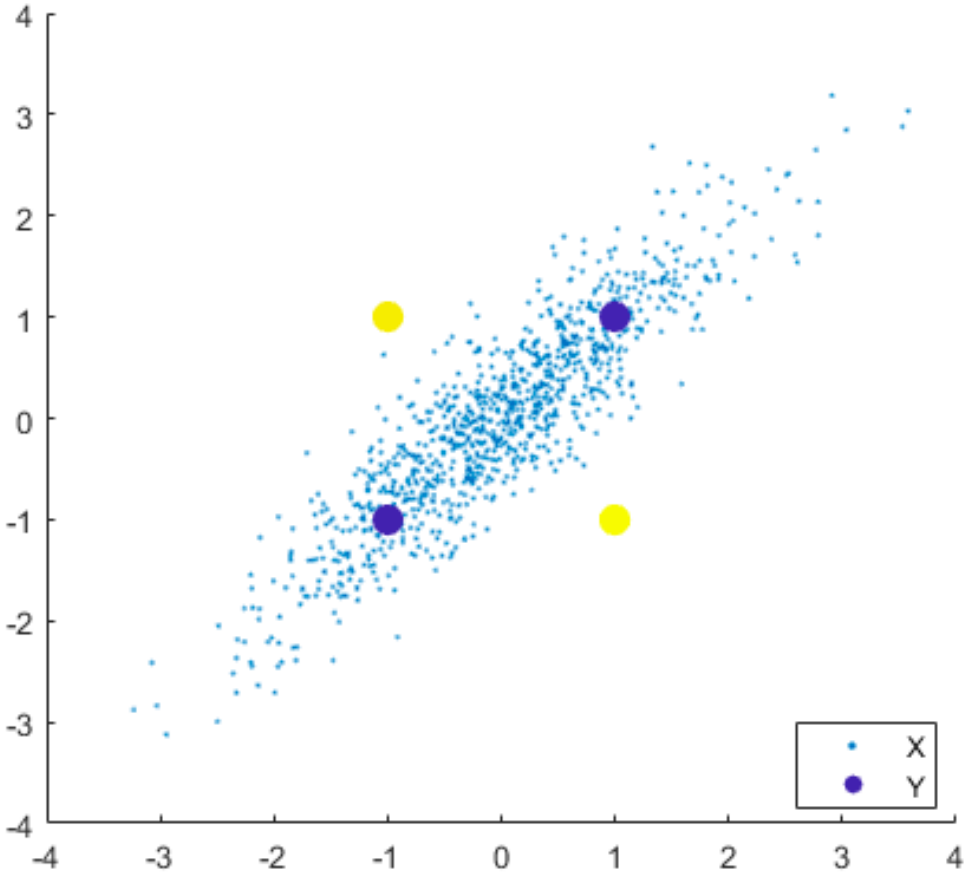
$$g_i(x) = -\frac{1}{2}(x-\mu_i)^t\Sigma_i^{-1}(x-\mu_i) - \frac{d}{2}\ln 2\pi - \frac{1}{2}\ln|\Sigma_i| + \ln P(\omega_i)$$

- When $P(\omega_i)$ is the same for each of the c classes

- Mahalanobis distance classifier

$$g_i(x) = -\frac{1}{2}(x-\mu_i)^t\Sigma_i^{-1}(x-\mu_i)$$

Mahalanobis Distance



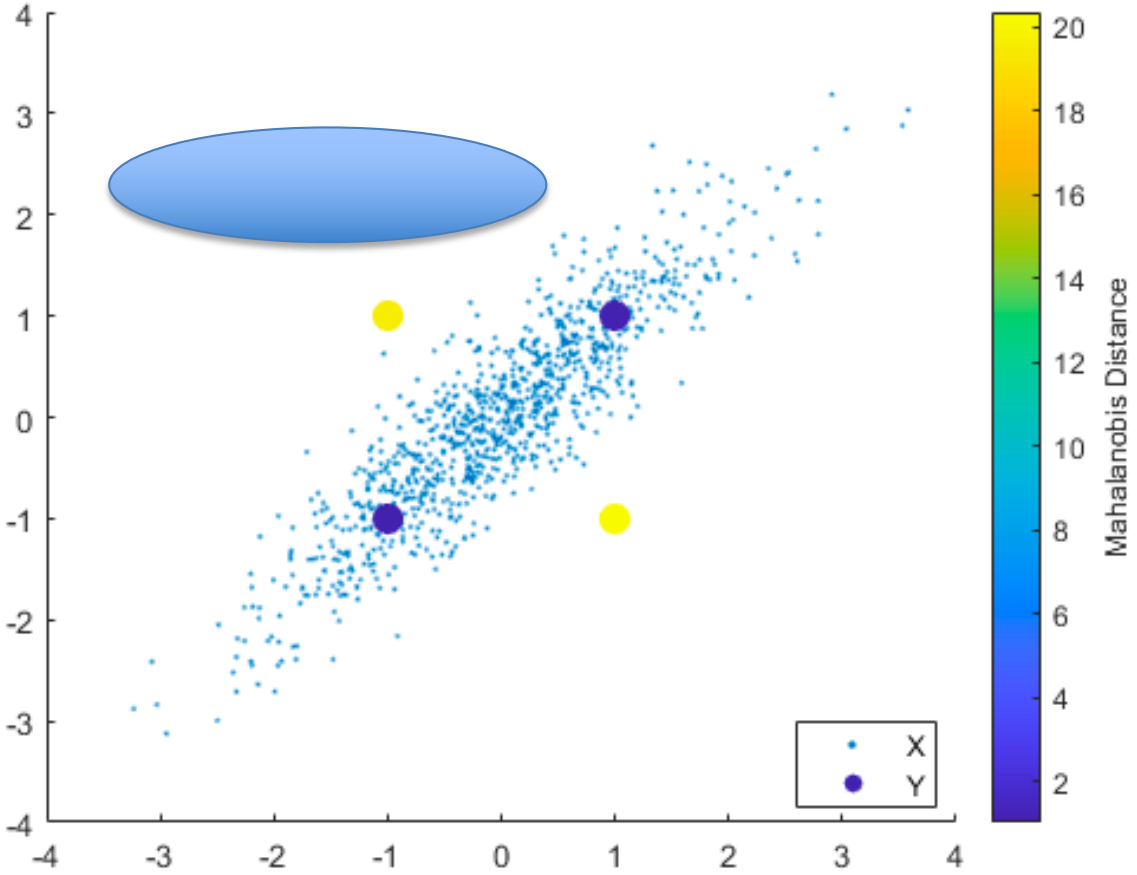
Euclidean Distance:

$$\mathrm{d} = \sqrt{\sum_{i=1}^{n} (X_i - Y_i)^2}$$

Mahalanobis Distance:

$$d\left(\vec{x}, \vec{y}\right) = \sqrt{\left(\vec{x} - \vec{y}\right)^{\top} S^{-1} \left(\vec{x} - \vec{y}\right)}$$

Mahalanobis Distance



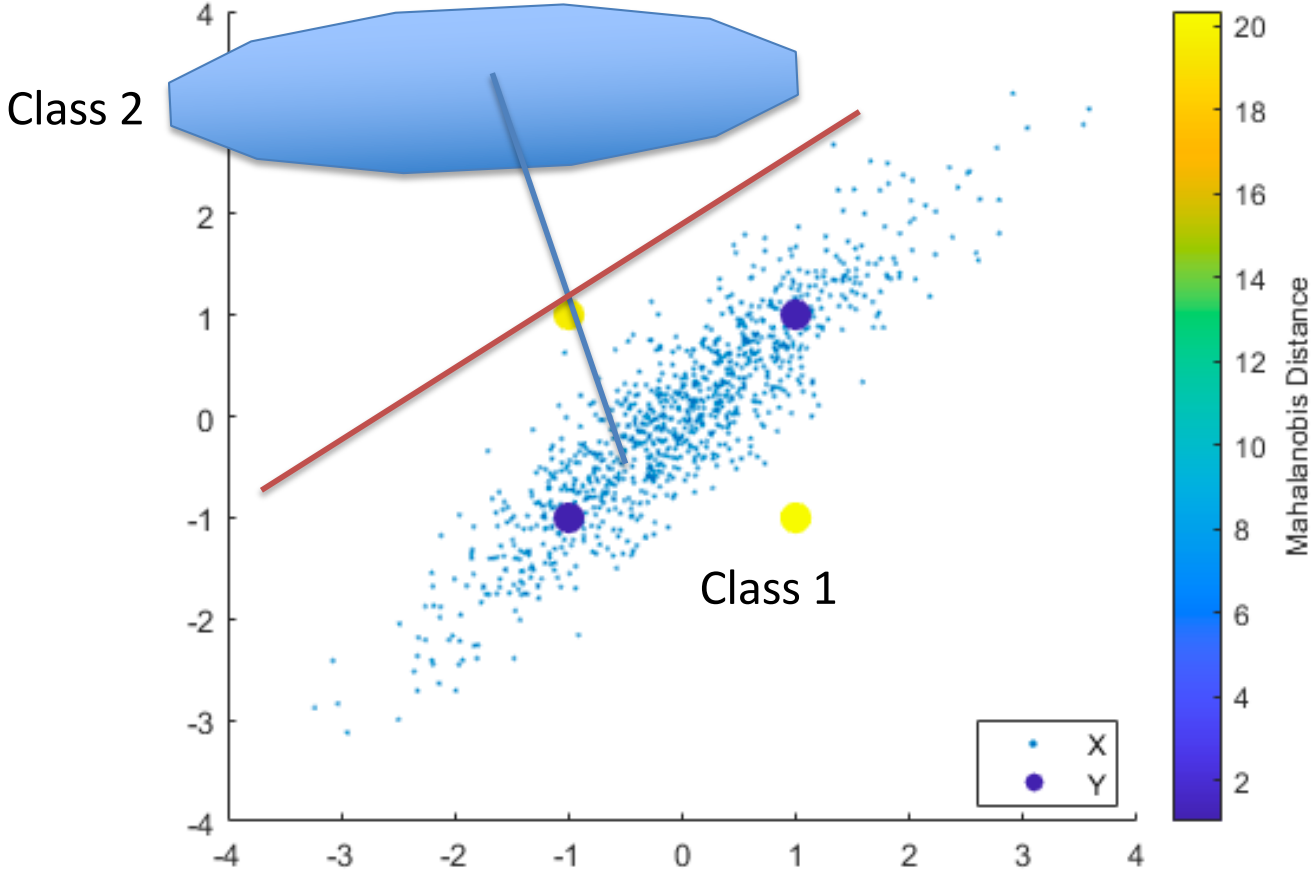
$$d = \sqrt{\sum_{i=1}^{n} (X_i - Y_i)^2}$$

Euclidean Distance:

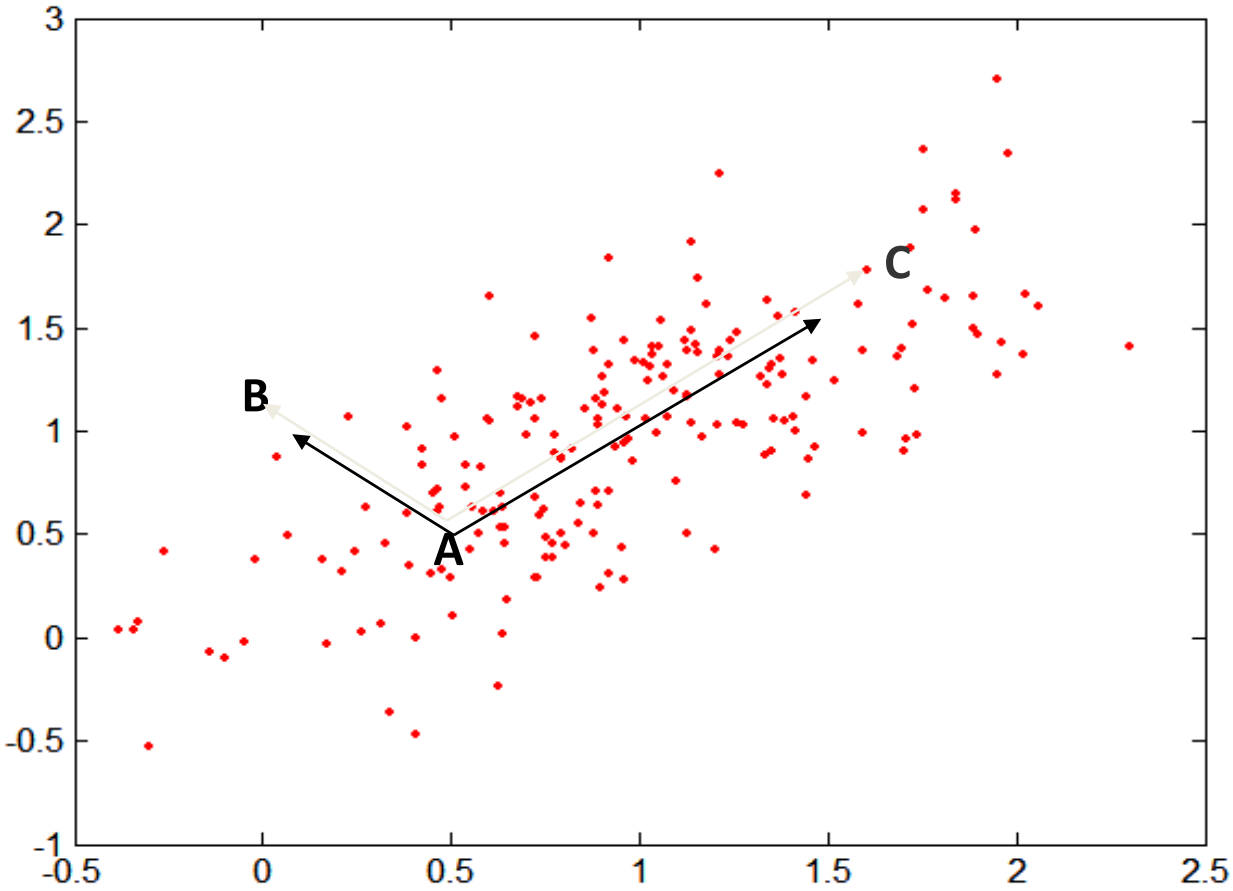
All points are equidistant

$$d\left(\vec{x}, \vec{y}\right) = \sqrt{\left(\vec{x} - \vec{y}\right)^{\top} S^{-1} \left(\vec{x} - \vec{y}\right)}$$

Mahalanobis Distance



Mahalanobis Distance



Covariance Matrix:

$$\Sigma = \begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.3 \end{bmatrix}$$

A: (0.5, 0.5)

B: (0, 1)

C: (1.5, 1.5)

Compute squared versions

Euclid(A,B)

Euclid(A,C)

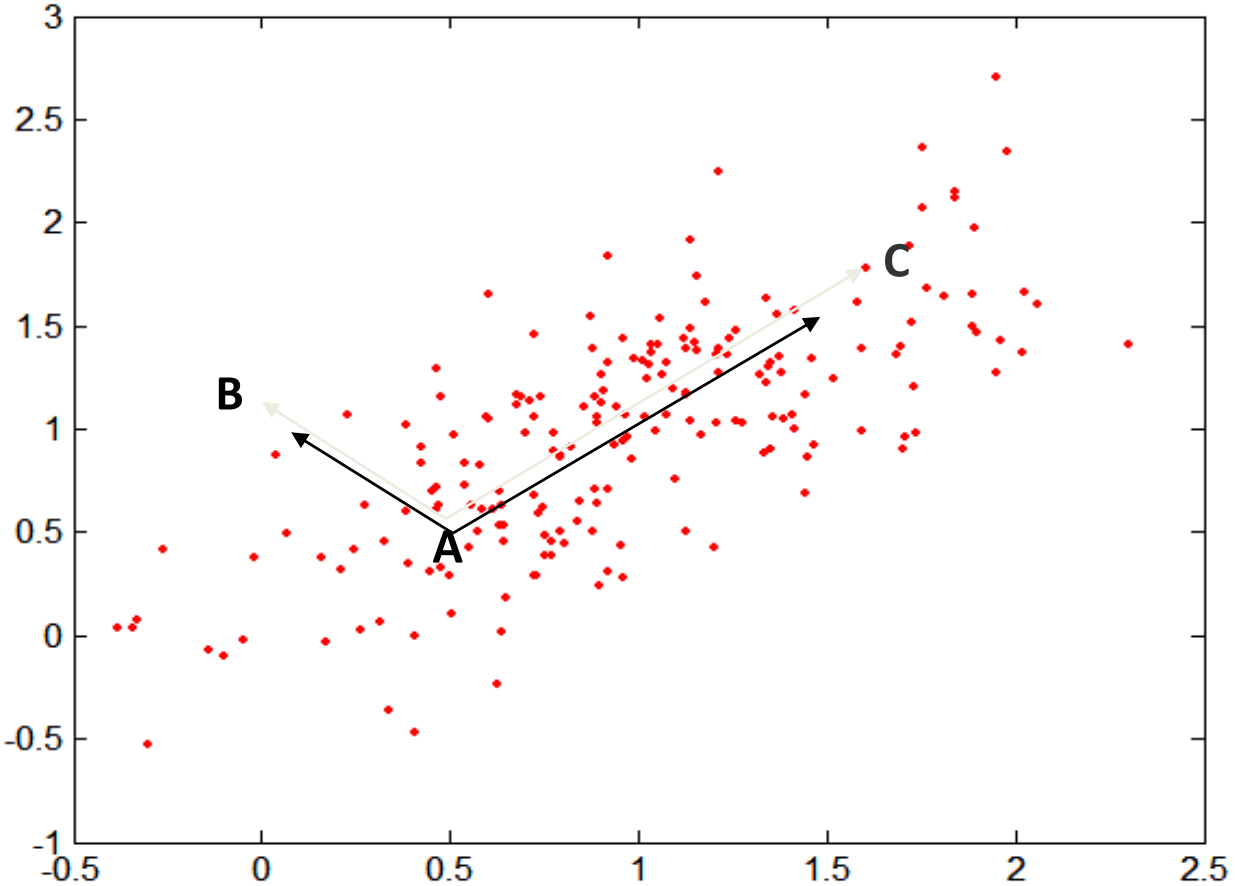
Mahal(A,B)

Mahal(A,C)

$$d = \sqrt{\sum_{i=1}^{n} (X_i - Y_i)^2}$$

$$d(\vec{x}, \vec{y}) = \sqrt{(\vec{x} - \vec{y})^\top S^{-1} (\vec{x} - \vec{y})}$$

Mahalanobis Distance



Covariance Matrix:

$$\Sigma = \begin{bmatrix} 0.3 & 0.2 \\ 0.2 & 0.3 \end{bmatrix}$$

A: (0.5, 0.5)

B: (0, 1)

C: (1.5, 1.5)

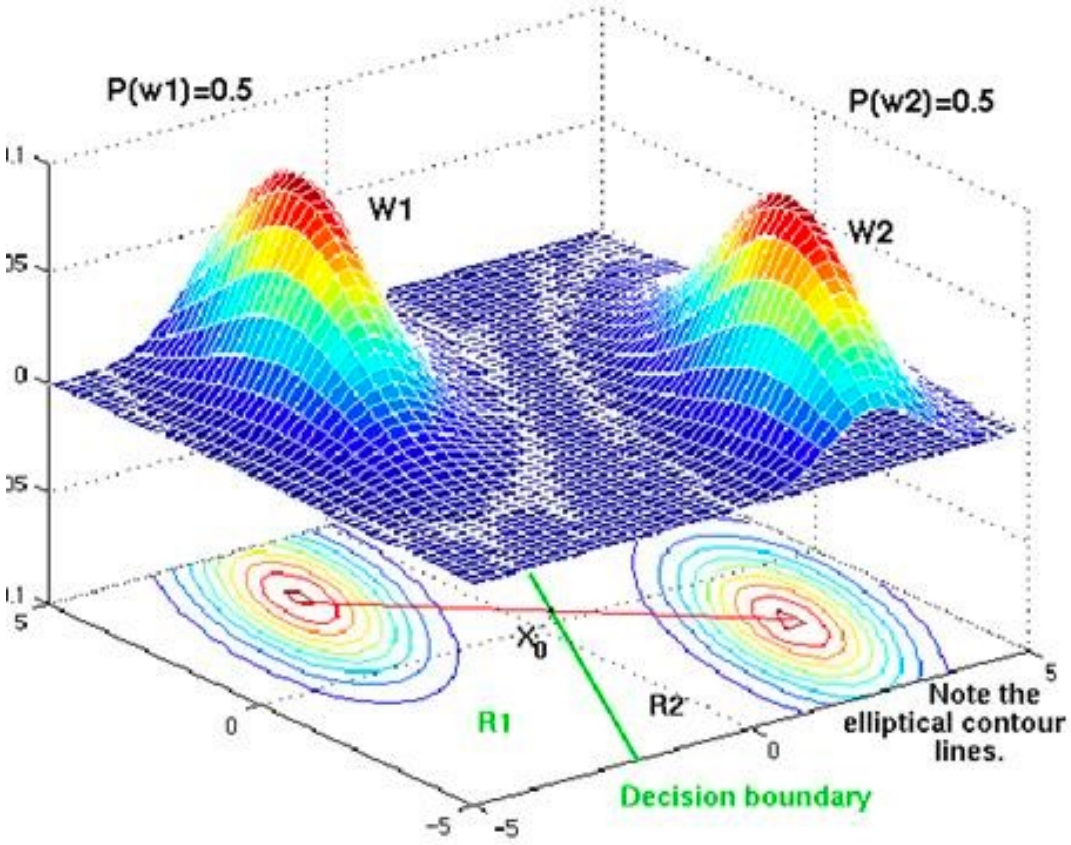
Euclid(A,B) = 0.5

Euclid(A,C) = 2

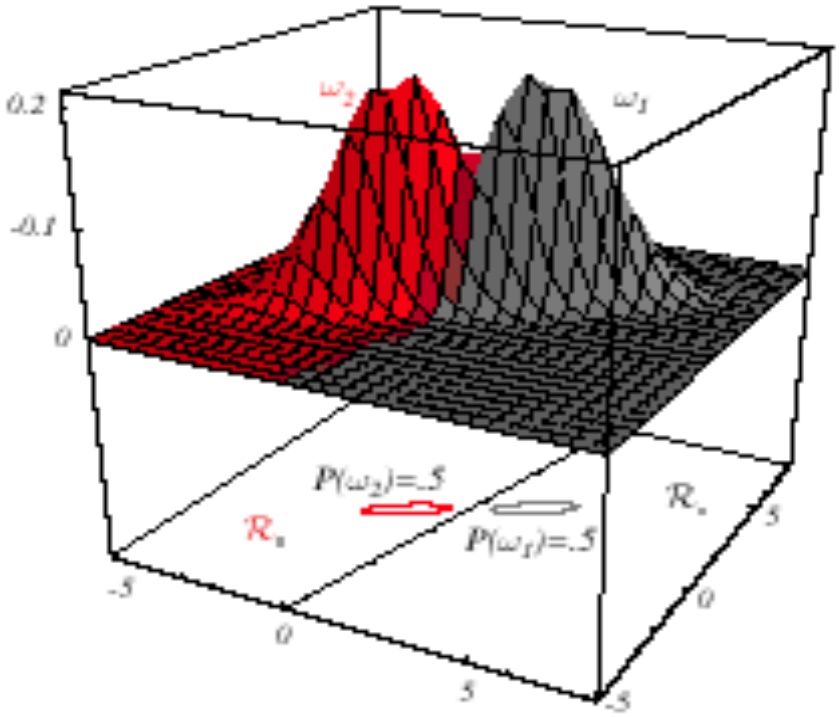
Mahal(A,B) = 5

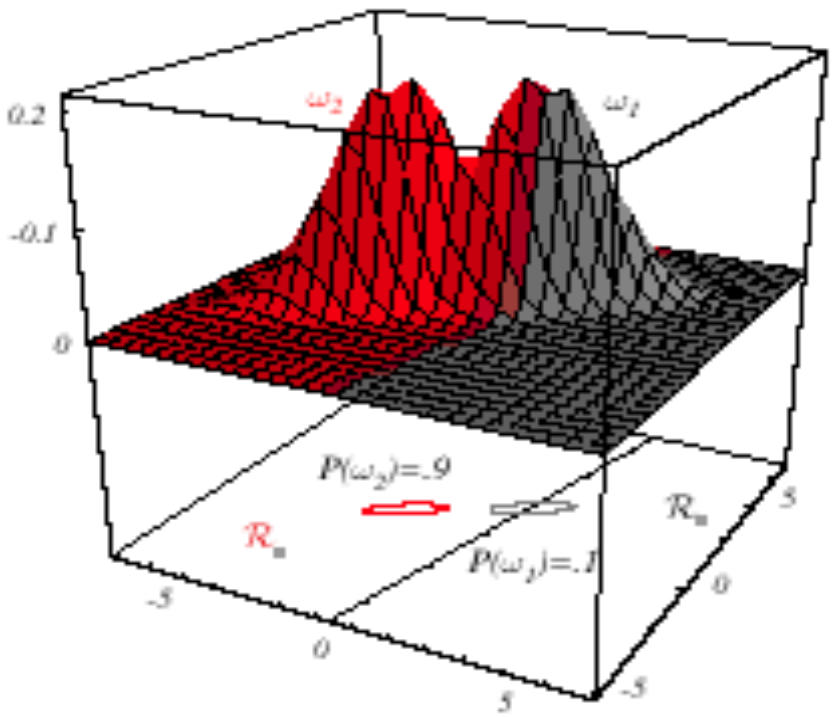
Mahal(A,C) = 4

Discriminant Functions for the Normal Density…



The contour lines are elliptical in shape because the covariance matrix is not diagonal. However, both densities show the same elliptical shape. The prior probabilities are the same, and so the point x0 lies halfway between the 2 means. The decision boundary is not orthogonal to the red line. Instead, it is tilted so that its points are of equal distance to the contour lines in w1 and those in w2.


0.2
-0.1
0
ω2
ω1
P(ω2)=.5
P(ω1)=.5
R1
R2
-5
0
5
-5
0
5


0.2
-0.1
0
ω2
ω1
P(ω2)=.9
P(ω1)=.1
R1
R2
-5
0
5
-5
0
5

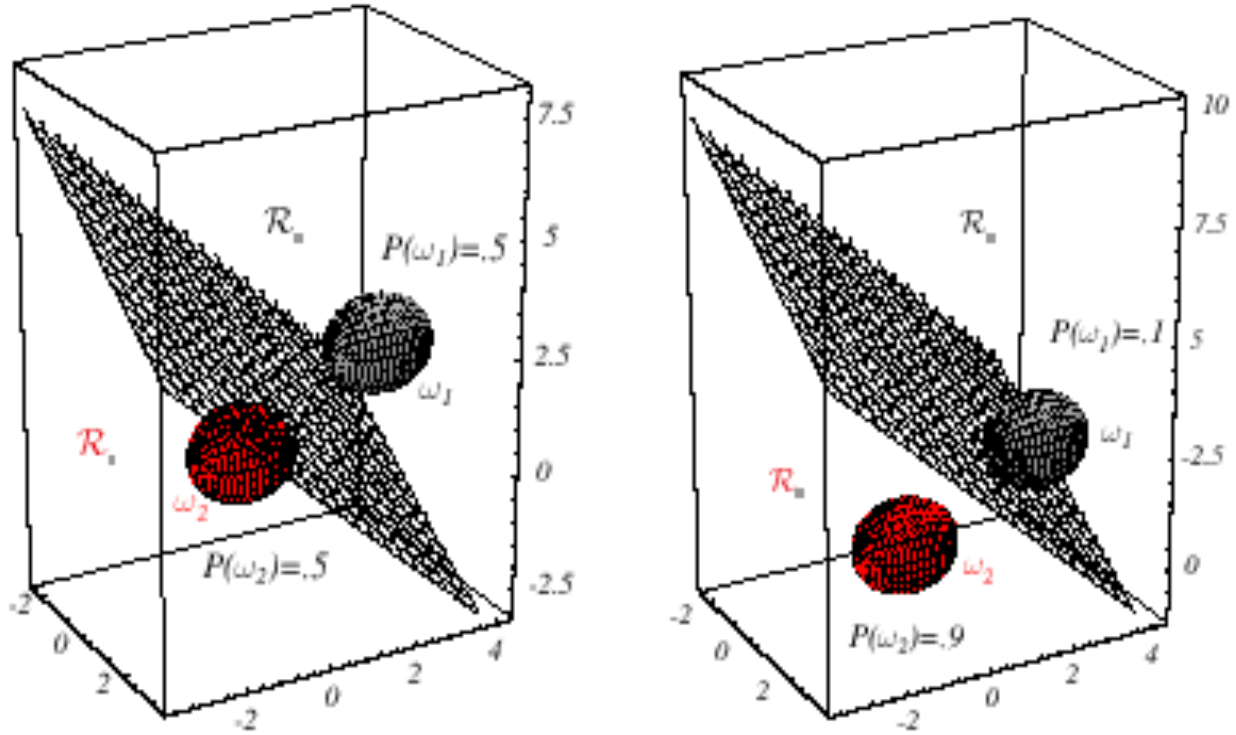


FIGURE 2.12. Probability densities (indicated by the surfaces in two dimensions and ellipsoidal surfaces in three dimensions) and decision regions for equal but asymmetric Gaussian distributions. The decision hyperplanes need not be perpendicular to the line connecting the means. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Discriminant Functions for the Normal Density…

- Case Σi = arbitrary
  - The covariance matrices are different for each category

$$g_i(x) = x^t W_i x + w_i^t x = w_{i0}$$

where :

$$W_i = -\frac{1}{2}\Sigma_i^{-1}$$

$$w_i = \Sigma_i^{-1}\mu_i$$

$$w_{i0} = -\frac{1}{2}\mu_i^t \Sigma_i^{-1}\mu_i - \frac{1}{2}ln\left|\Sigma_i\right| + ln\,P(\omega_i)$$

Discriminant Functions for the Normal Density…

Disconnected decision regions

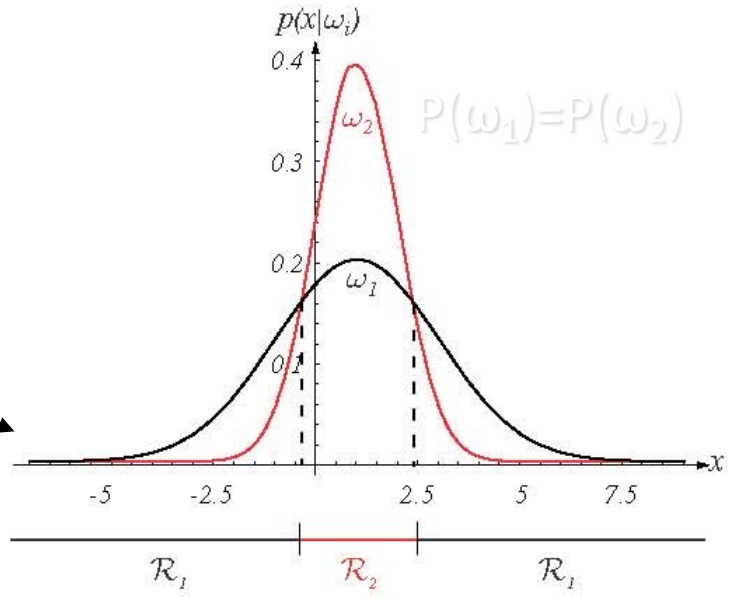
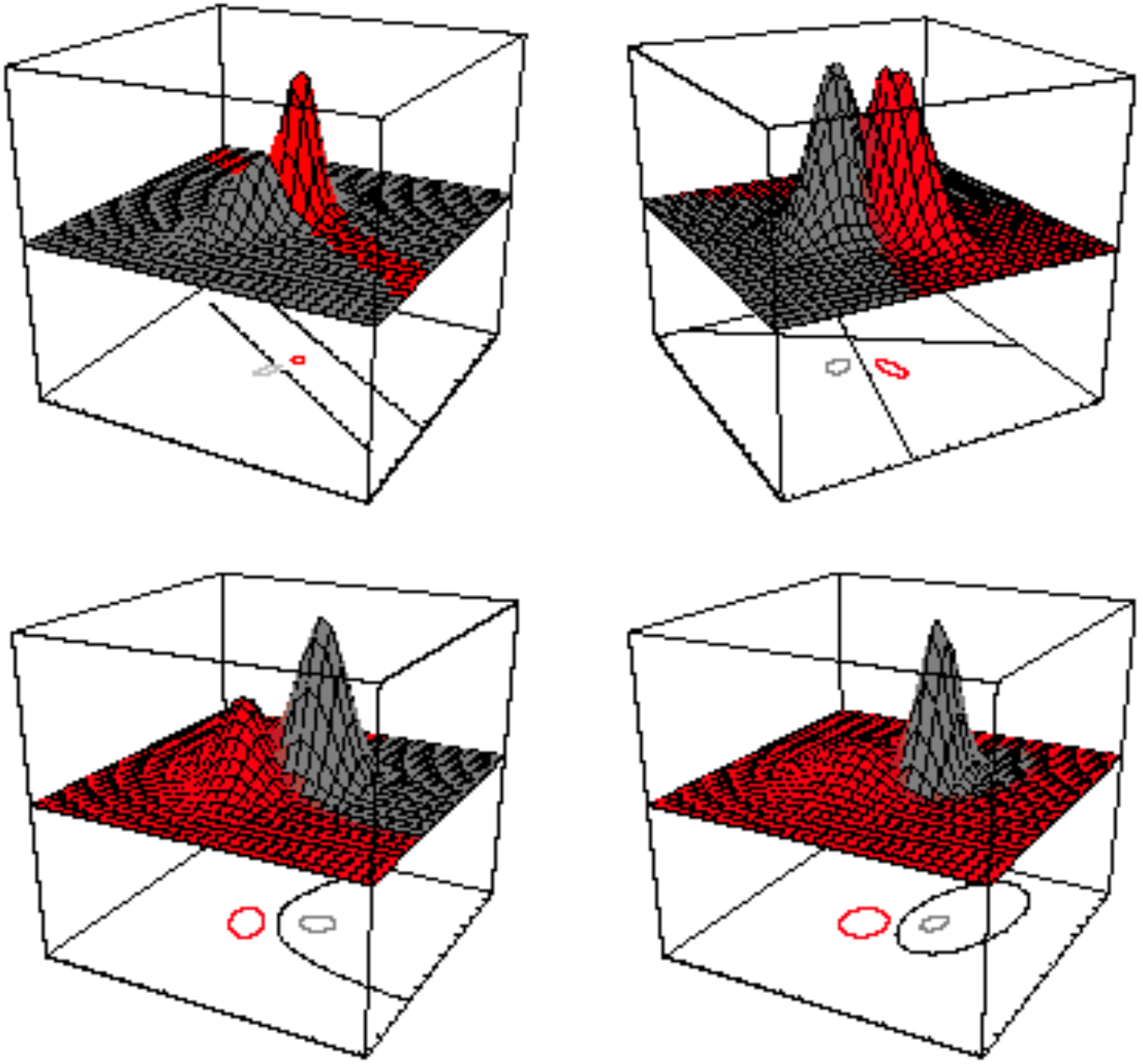


FIGURE 2.13. Non-simply connected decision regions can arise in one dimensions for Gaussians having unequal variance. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.



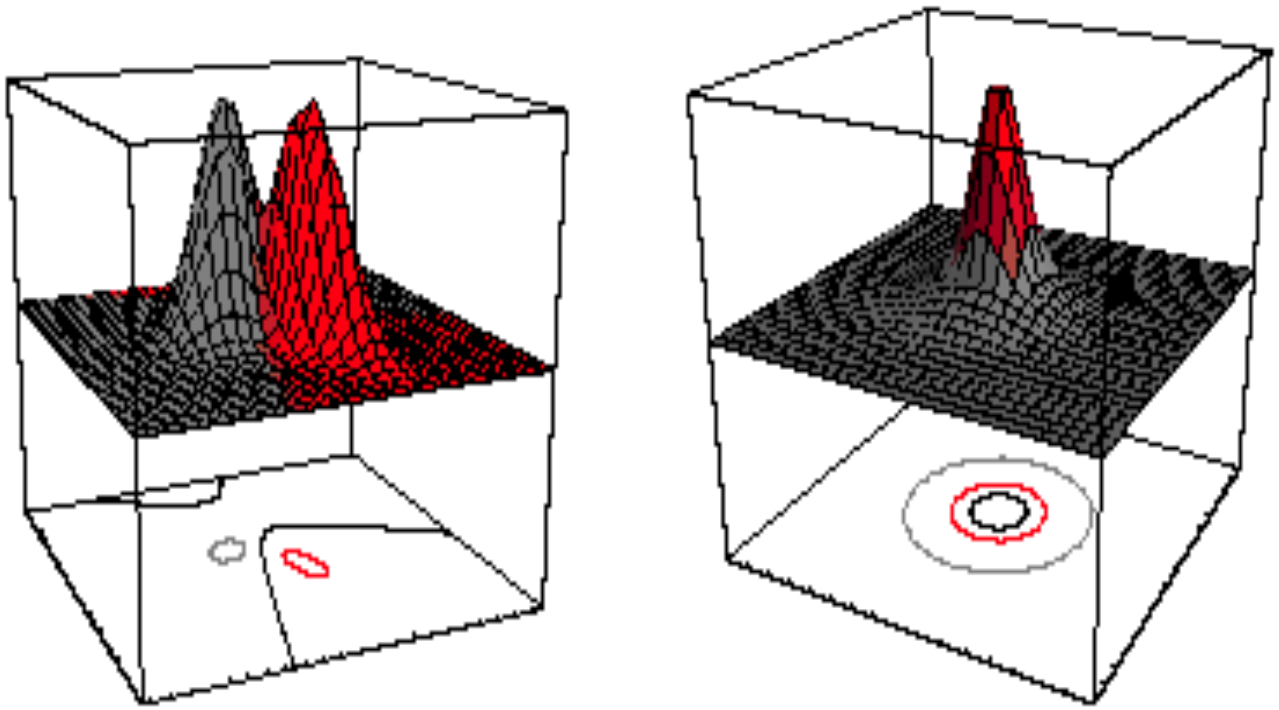


FIGURE 2.14. Arbitrary Gaussian distributions lead to Bayes decision boundaries that are general hyperquadrics. Conversely, given any hyperquadric, one can find two Gaussian distributions whose Bayes decision boundary is that hyperquadric. These variances are indicated by the contours of constant probability density. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Discriminant Functions for the Normal Density

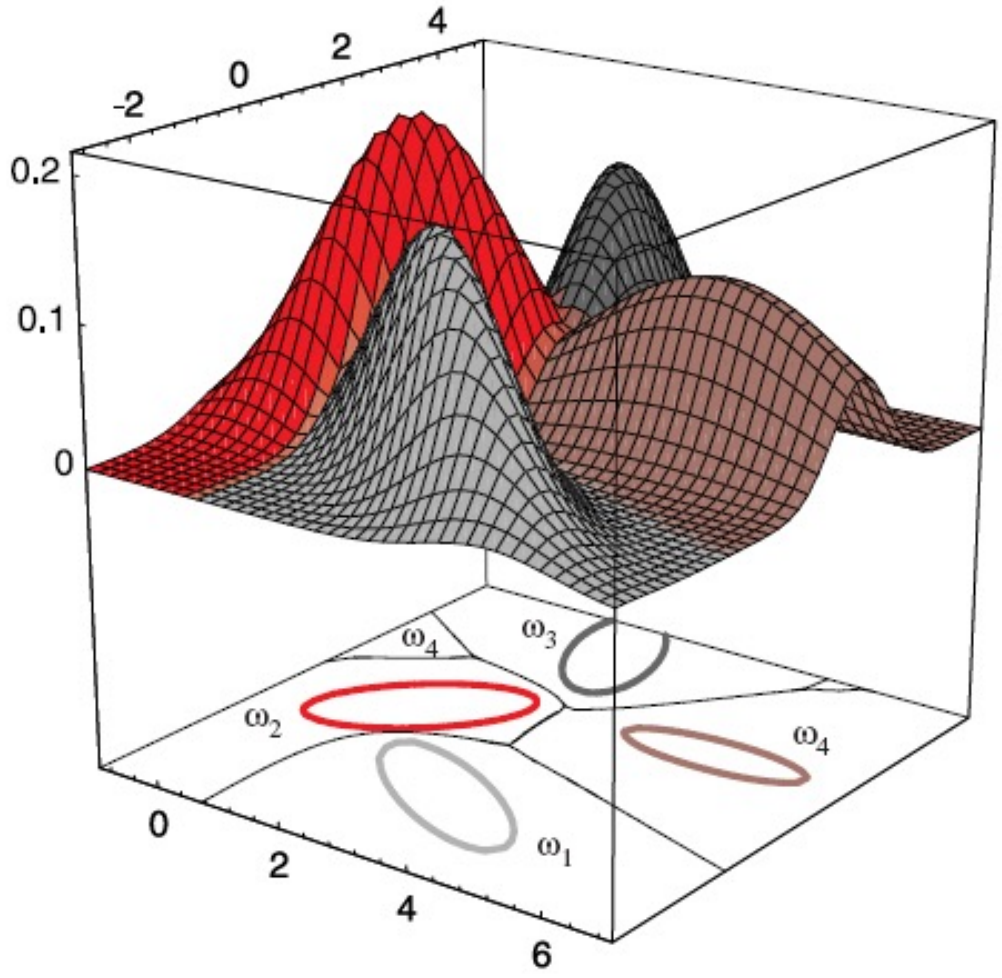
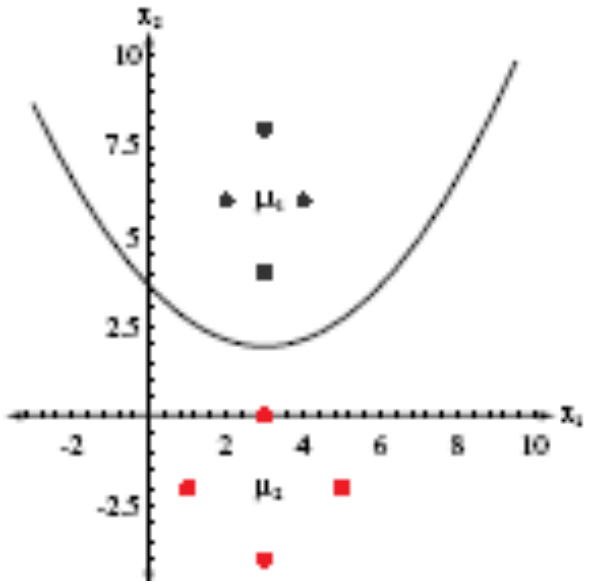


Figure 2.16: The decision regions for four normal distributions. Even with such a low number of categories, the shapes of the boundary regions can be rather complex.

Decision Regions for Two-Dimensional Gaussian Data

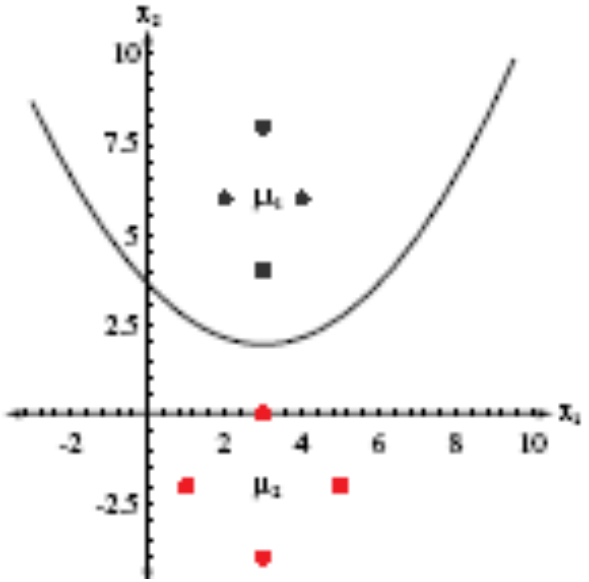


$$\mu_1 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}; \quad \Sigma_1 = \begin{pmatrix} 1/2 & 0 \\ 0 & 2 \end{pmatrix} \text{ and } \mu_2 = \begin{bmatrix} 3 \\ -2 \end{bmatrix}; \quad \Sigma_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

$$\Sigma_1^{-1} = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix} \quad \text{and} \quad \Sigma_2^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

$$P(\omega_1) = P(\omega_2) = 0.5$$

Decision Regions for Two-Dimensional Gaussian Data



$$\mu_1 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}; \quad \Sigma_1 = \begin{pmatrix} 1/2 & 0 \\ 0 & 2 \end{pmatrix} \text{ and } \mu_2 = \begin{bmatrix} 3 \\ -2 \end{bmatrix}; \quad \Sigma_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

$$\Sigma_1^{-1} = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix} \quad \text{and} \quad \Sigma_2^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

$$P(\omega_1) = P(\omega_2) = 0.5$$

$$x_2 = 3.514 - 1.125x_1 + 0.1875x_1^2$$

Thanks.